Response-aware path compression

Sharp planar distortion and exact Euler storage

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Abstract

A Hilbert–Schmidt signature certificate can reject a representation whose worst-case linear-system response error is acceptable. For ordinary planar bounded-variation paths, degree-two log-ODE distortion has sharp small-scale response/certificate ratio $2\sqrt{2}/9$, with explicit finite-scale and perturbation bounds. Homogeneous area chains give a separation at every positive scale. An exact bounded-variation Euler minimax constraint changes finite integer storage optima while preserving their leading asymptotics. Proofs and reproducible examples are included.

Discussion draft. Arguments and computational checks were prepared with AI assistance. Independent expert validation and confirmation of novelty remain outstanding.

1 Criterion and comparison

The response criterion of Ferrucci, Perrée and Lyons (FPL) [1] motivates retaining a path's effect on bounded linear systems while reducing storage. Fix d driver channels and n response states. Real matrix tuples $A = (A_1, \dots, A_d)$ have joint budget

$$|A|^2 = \sum_i \|A_i\|_F^2 \le r^2.$$

Tensor words are evaluated chronologically: $R_A(e_{i_1} \otimes \cdots \otimes e_{i_j}) = A_{i_1} \cdots A_{i_j}$. This is the row-state convention $dY = \sum_i Y A_i \, dX^i$; transpose tuples to recover column states. Starting at identity, $\Phi^A(X) = R_A(S(X))$ is the terminal response operator. For full tensor series F, G, put

$$\Delta_{r,n}(F,G) = \sup_{|A|\le r} \|R_A(F) - R_A(G)\|_{\mathrm{op}}, \qquad \mathcal{B}_r(F,G) = \sum_{j\ge 1} r^j \, \|F_j - G_j\|_{\mathrm{HS}}. \tag{1}$$

Here operator norm is Euclidean, and standard word tensors are orthonormal for HS norm. In the classes below these series converge absolutely and $\Delta_{r,n} \le \mathcal{B}_r$. Full HS certificate, a further length-based envelope, and true response error are different constraints. Straight scalar developments can attain their HS certificate; a restricted Lie-direction gap is compatible with that fact.

2 Planar bounded-variation distortion

Let $X : [0,1] \to \mathbb{R}^2$ be continuous with finite Euclidean variation L. Fix a partition π; X_i is the restriction to interval i. Write $\ell_{i,j} = (\log_\otimes S(X_i))_j$ for its degree-j local log-signature, and define

$$F = S(X), \qquad G_\pi = \prod_i \exp_\otimes(\ell_{i,1} + \ell_{i,2}), \qquad D_3 = \sum_i \ell_{i,3}, \quad h = \|D_3\|_{\mathrm{HS}}.$$

Dilation δ_s multiplies tensor degree j by s^j. Coefficient radius r and path scale s combine into $t = sr$. Set $\Delta_n(t) = \Delta_{1,n}(\delta_t F, \delta_t G_\pi)$ and $\mathcal{B}(t) = \mathcal{B}_1(\delta_t F, \delta_t G_\pi)$. Homogeneity gives $\Delta_{r,n}(\delta_s F, \delta_s G_\pi) = \Delta_n(sr)$.

Theorem 1. *If $h > 0$, then for every $n \geq 2$,*

$$\frac{\Delta_n(t)}{\mathcal{B}(t)} \longrightarrow \rho := \frac{2\sqrt{2}}{9} \quad (t \downarrow 0),$$

uniformly in n. In particular, the asymptotic HS/response overestimation factor is $1/\rho = 9/(2\sqrt{2})$, approximately 3.18. Explicit tails below give a positive finite-scale separation radius and uniformity on a BV perturbation neighborhood.

Put $a = \sum_i \|\ell_{i,1}\|_2$, $b = \sum_i \|\ell_{i,2}\|_{\mathrm{HS}}$ and

$$T_4(a, b; t) = e^{at+bt^2} - 1 - at - (a^2/2 + b)t^2 - (a^3/6 + ab)t^3.$$

All coefficients in this degree-four-and-above tail are nonnegative. Define $Q_R = T_4(L, 0; t) + T_4(a, b/2; t)$ and $Q_H = T_4(L, 0; t) + T_4(a, b; t)$. Then

$$|\Delta_n(t) - \rho h t^3| \leq Q_R(t), \qquad ht^3 \leq \mathcal{B}(t) \leq ht^3 + Q_H(t). \tag{2}$$

For any reference $T > 0$, let $C_T = Q_R(T)/T^4$ and

$$t_* = \min\{T, (1 - \rho)h/(2C_T)\}, \qquad c = (1 + \rho)/2. \tag{3}$$

For $0 < t \leq t_*$, $\Delta_n(t) \leq cht^3 < ht^3 \leq \mathcal{B}(t)$. Appendix A proves these assertions and perturbation stability.

A finite decision with exact arithmetic. For $0 \to e_1 \to e_1 + e_2$ on one interval,

$$D_3 = ([e_1, Z] - [e_2, Z])/12, \quad Z = [e_1, e_2], \quad h = \sqrt{3}/6, \quad a = \sqrt{2}, \quad b = \sqrt{2}/2.$$

At $t = 1/20$, rational Taylor enclosures prove

$$\Delta_n(t) < 1.934\,10^{-5} < 2\,10^{-5} < 3.608\,10^{-5} < \mathcal{B}(t). \tag{4}$$

Thus tolerance $2\,10^{-5}$ accepts this approximation under a proved response bound and rejects its full HS certificate. Appendix F gives the enclosure, without optimizer or floating-point rounding.

The partition and $h > 0$ are essential. A two-interval square loop has zero cubic discrepancy but nonzero quartic error. Mesh refinement can make h vanish. Theorem 1 supplies no mesh-uniform conclusion or extension to general rough input with $2 < p < 3$.

3 Homogeneous merges at every positive scale

For $\ell \geq 3$, in driver dimension $d = \ell + 1$, let $U_i = [e_i, e_{i+1}]$ and compare decoded depth-two series

$$F_\lambda = \prod_{i=1}^{\ell} e_\otimes^{\lambda U_i}, \qquad G_\lambda = e_\otimes^{\lambda \sum_i U_i}, \qquad \lambda > 0.$$

Use unit tuple budget in $\Delta_n(\lambda) = \Delta_{1,n}(F_\lambda, G_\lambda)$ and $\mathcal{B}(\lambda) = \mathcal{B}_1(F_\lambda, G_\lambda)$.

Theorem 2. *For every $\ell \geq 3$, every response dimension $n \geq 1$ and $\lambda > 0$,*

$$\Delta_n(\lambda) < \frac{\lambda^2}{\sqrt{2}} e^{\sqrt{2}\lambda} < \frac{e^{\sqrt{6}\lambda} - 1 - \sqrt{6}\lambda}{\sqrt{3}} \leq \mathcal{B}(\lambda). \tag{5}$$

The separation bounds are independent of chain length.

This permits an ℓ-to-one merge at a nonempty interval of response tolerances rejected by the full certificate. With generic depth-two log-signature layout held fixed, storage falls by factor ℓ. The result concerns the displayed decoded factors. For ordinary coordinate loops, add the fine decoder's original-path error to assess total error. Equivalently, the fine series is exact for a concatenation of pure area rough blocks. Appendix B verifies the infinitely many coefficient comparisons analytically; a finite coefficient scan would be insufficient.

4 Exact Euler minimax and storage

For degree-N Euler decoding on m intervals, put $P_N(x) = \sum_{j=0}^{N} x^j/j!$, and

$$E^A_{N,\pi}(X) = \prod_i \left[I + \sum_{j=1}^{N} R_{A,j}(S(X_i)_j) \right].$$

Theorem 3. *For any $N, m \geq 1$, fixed driver and response dimensions, coefficient radius $r > 0$, and paths of variation at most one,*

$$\sup_X \inf_{\pi:\, m \text{ intervals}} \sup_{|A|\leq r} \left\| \Phi^A(X) - E^A_{N,\pi}(X) \right\|_{\mathrm{op}} = \mathcal{E}(N, m, r) := e^r - P_N(r/m)^m. \tag{6}$$

Partitions may depend on X, but not on A. Equal-variation partitions give the upper bound; a straight line with a scalar/rank-one development attains the minimax error.

FPL already display the scalar expression in their equation (1.13). Appendix C proves its universal majorant and partition minimization. This refines the additional length envelope

$$B(N, m, r) = \frac{e^r r^{N+1}}{(N+1)! m^N}.$$

Both comparisons use the same dense generic storage $m\Lambda_d(N)$, where

$$\Lambda_d(N) = \sum_{j=1}^{N} \frac{1}{j} \sum_{v|j} \mu(v) d^{j/v};$$

μ is the Möbius function. The decoder is Euler even though stored coordinates are log-signature coordinates.

Finite integer movement. For $d = 2, r = 1, \varepsilon = 10^{-2}$, the two optima are

Constraint	N	m	stored scalars
Exact BV minimax	4	1	8
Length envelope	5	1	14

Appendix F certifies global minimality with rational arithmetic. On the retained grid $d \in \{2, 3, 5\}$, $r \in \{0.1, 0.5, 1, 2, 5, 10, 20, 50, 100\}$ and $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-4}, 10^{-8}, 10^{-12}\}$, optimum sets change in 85 of 135 cases; 84 have strictly lower storage, with largest saving 42.9%. These are finite-grid observations. Integer searches use high-precision boundary refinement; the general grid is not interval certified.

Leading asymptotics persist. Let $C_{d,k}(z, \varepsilon)$ minimize $m\Lambda_d(kq)$ under $\mathcal{E}(q, m, z) \leq \varepsilon$, with fixed $d \geq 2, k \geq 1$. Put $\beta = k \log d$, $\lambda = z + \log(z/\varepsilon)$. In either limit $z \to \infty$ at fixed $\varepsilon > 0$, or $\varepsilon \downarrow 0$ at fixed $z > 0$, any optimizer satisfies

$$q_* \sim \sqrt{\lambda/\beta}, \qquad \log(m_*/z) \sim \sqrt{\beta\lambda}, \qquad \log(C_{d,k}/z) \sim 2\sqrt{\beta\lambda}. \tag{7}$$

Replacing $\mathcal{E}$ by B changes minimum cost by a relative factor tending to one (Appendix D). For $k = 1$, this preserves the leading BV storage asymptotics in FPL.

Pure Engel paths supply a separate restricted comparison, with response dimension $n \geq 2$. For $k \geq 2$ and $E_k = \mathrm{ad}_{e_1}^{k-1}(e_2)$, set

$$h_k = \sqrt{\binom{2k-2}{k-1}}, \quad g_k = [2(k-1)]^{(k-1)/2} k^{-k/2}, \quad s_k = g_k/\sqrt{2}.$$

For a pure degree-k rough path, take $p \geq k$; effective amplitude $a = br^k > 0$ absorbs original amplitude b and coefficient radius r. When $\varepsilon < e^{as_k} - 1$, true Euler storage is bracketed by

$$C_{d,k}(as_k, \varepsilon) \leq C_k^{\mathrm{true}}(a, \varepsilon) \leq C_{d,k}(ag_k, \varepsilon) \leq C_{d,k}(ah_k, \varepsilon).$$

Consequently, at fixed d, k, ε,

$$\liminf_{a \to \infty} \frac{1}{\sqrt{a}} \log \frac{C_{d,k}(ah_k, \varepsilon)}{C_k^{\mathrm{true}}(a, \varepsilon)} \geq 2\sqrt{k \log d}\,(\sqrt{h_k} - \sqrt{g_k}) > 0.$$

Appendix E gives the gauges, witness and scope. A single depth-k log-ODE factor represents this pure path exactly; the growing overhead concerns fixed Euler decoding and dense generic storage.

5 Prior work and boundaries

Development separation through polynomial identities is already explicit in Chevyrev–Lyons, §4.2 and Theorem 4.8 [2]. The planar degree-six matrix identity is classical [3]; low-degree dimension-two attainment implies no universal dimension sufficiency. Spectral Lie-polynomial evaluations and low-degree calculations have direct precedent in Boedihardjo–Geng–Souris [4], with different norm budgets; exact pure-path tail asymptotics appear in Li–Gao [5]. The Engel gauge below is a Böttcher–Wenzel corollary [6]. The product-formula representation is standard [7], as is fixed-system adaptive log-ODE analysis [8]. FPL's discussion after (2.10) already identifies the first discarded local log-signature error.

The candidate addition is the constrained response/HS comparison and its finite-scale consequences. General rough inputs with $2 < p < 3$, mesh-uniform nondegeneracy, exact finite-scale response suprema and general cubic-zero behavior remain open. The displayed results give no correction to FPL's general rough-path asymptotics or its empirical averaged-error figures.

A Planar gauge, tails and stability

Signature degree j is $S(X)_j = \int_{0<u_1<\cdots<u_j<1} dX_{u_1} \otimes \cdots \otimes dX_{u_j}$. For homogeneous D of degree j, write $\gamma_{j,n}(D) = \sup_{|A|\le 1} \|R_A(D)\|_{\mathrm{op}}$. Let $Z = [e_1, e_2]$. Every planar cubic Lie element is $D = [w, Z]$ with $w \in \mathbb{R}^2$, and $\|D\|_{\mathrm{HS}} = \sqrt{6}\,\|w\|_2$. Proper driver rotations preserve the tuple ball and fix Z, so reduce to $D = \|w\|_2\,[e_1, Z]$. Böttcher–Wenzel and scalar maximization give

$$\|[A_1, [A_1, A_2]]\|_{\mathrm{op}} \le 2\,\|A_1\|_F^2\,\|A_2\|_F \le \frac{4}{3\sqrt{3}}.$$

Equality holds at

$$A_1 = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad A_2 = \frac{1}{\sqrt{3}}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

whose cubic response is $4E_{12}/(3\sqrt{3})$. Embedding proves equality for every $n \ge 2$; hence $\gamma_{3,n}(D) = \rho\,\|D\|_{\mathrm{HS}}$. One commutator step similarly gives $\gamma_{2,n}(\alpha Z) = |\alpha|/\sqrt{2} = \|\alpha Z\|_{\mathrm{HS}}/2$. It is attained at $A_1 = \mathrm{diag}(1,-1)/2$ and $A_2 = E_{12}/\sqrt{2}$.

First discrepancy. Group-like series F, G agreeing through degree two have $D_3 = F_3 - G_3 = (\log F)_3 - (\log G)_3$, a cubic Lie tensor. The cubic formula and uniform response tail yield

$$\rho h t^3 - Q_R(t) \le \Delta_n(t) \le \rho h t^3 + Q_R(t).$$

The certificate includes its degree-three term exactly, giving (2). Thus

$$\left|\frac{\Delta_n(t)}{\mathcal{B}(t)} - \rho\right| \le \frac{Q_R(t) + \rho Q_H(t)}{h t^3}.$$

Both tails are $O(t^4)$, independently of n. For partitioned degree-two log-ODE, interval factors agree through degree two and first differ by $\ell_{i,3}$. Multiplication by any nonconstant factor raises degree, so $D_3 = \sum_i \ell_{i,3}$.

Full series control. BV factorial bounds majorize F by e^{Lt}. Tensor products majorize G_π by e^{at+bt^2}. The sharp area gauge majorizes the matrix decoder by $e^{at+(b/2)t^2}$. Removing degrees zero to three yields Q_R, Q_H. For numerical evaluation, the positive identity

$$T_4(a,b;t) = e^p - P_3(p) + q(e^p - P_1(p)) + e^p(e^q - P_1(q)), \quad p = at,\ q = bt^2,$$

avoids small-scale subtractive cancellation. Nonnegative coefficients imply $Q_R(t) \le C_T t^4$ on $0 \le t \le T$, which proves (3) and the strict gap.

BV stability. For anchored paths of length at most L and $\eta = \|X - Y\|_{1\text{-var}}$, interpolation and insertion of one $d(Y - X)$ into the iterated integral give

$$\|S_j(X) - S_j(Y)\|_{\mathrm{HS}} \le \frac{\eta L^{j-1}}{(j-1)!}.$$

At the insertion time, summing over positions produces $(L_{\mathrm{before}} + L_{\mathrm{after}})^{j-1}/(j-1)!$; integrate against the variation measure of $Y - X$, then along interpolation. In

$$\ell_3 = S_3 - \tfrac{1}{2}(S_1 \otimes S_2 + S_2 \otimes S_1) + \tfrac{1}{3}S_1^{\otimes 3},$$

the three differences have bounds $\eta L^2/2$, $3\eta L^2/2$, and ηL^2. Therefore $\|\ell_3(X) - \ell_3(Y)\|_{\mathrm{HS}} \le 3\eta L^2$. On a fixed common partition, local difference variations sum to η; the same bound applies to D_3. Also $a \le L$, $b \le \sum_i L_i^2 \le L^2$. For a baseline of length L_X and cubic norm $h_X > 0$, set

$$L = L_X + 1, \qquad \eta_* = \min\{1, h_X/(6L^2)\}.$$

Perturbations within η_* have cubic norm at least $h_X/2$. Use $Q_R^*(t) = T_4(L, 0; t) + T_4(L, L^2/2; t)$ and this lower norm in (3) for a common positive radius.

Mixed blocks. For increments v_i and areas $\alpha_i Z$, compare

$$e_{\otimes}^{tv_1+t^2\alpha_1 Z} e_{\otimes}^{tv_2+t^2\alpha_2 Z} \quad \text{with} \quad e_{\otimes}^{t(v_1+v_2)+t^2(\alpha_1+\alpha_2+\omega/2)Z}, \qquad \omega = \det(v_1, v_2).$$

BCH at degree three gives $[w, Z]$, where

$$w = \tfrac{1}{2}(\alpha_2 v_1 - \alpha_1 v_2) + \tfrac{\omega}{12}(v_1 - v_2).$$

The same lemma applies when $w \neq 0$, with response tails $T_4(a_f, b_f/2; t) + T_4(a_c, b_c/2; t)$, where

$$a_f = \|v_1\| + \|v_2\|, \quad b_f = \sqrt{2}(|\alpha_1| + |\alpha_2|), \quad a_c = \|v_1 + v_2\|, \quad b_c = \sqrt{2}|\alpha_1 + \alpha_2 + \omega/2|.$$

For independent v_1, v_2, $w = 0$ exactly when $\alpha_1 = \alpha_2 = -\omega/6$. Pure planar area blocks commute.

B Gram certificates and all-scale chain proof

Let same-layer blocks $U_1, \ldots, U_\ell$ have Gram matrix $g_{ij} = \langle U_i, U_j \rangle_{\mathrm{HS}}$. For a weak composition α of m, let $w(\alpha)$ be its nondecreasing word, $\alpha! = \prod_i \alpha_i!$, and $W_\alpha = U_1^{\otimes \alpha_1} \otimes \cdots \otimes U_\ell^{\otimes \alpha_\ell}$. The meta-degree terms are

$$F_m = \sum_{|\alpha|=m} W_\alpha/\alpha!, \qquad G_m = (\sum_i U_i)^{\otimes m}/m!.$$

With $H_m = m!\,\|F_m - G_m\|_{\mathrm{HS}}$, tensor inner products give

$$H_m^2 = \sum_{|\alpha|=|\beta|=m} \binom{m}{\alpha}\binom{m}{\beta} \prod_{v=1}^{m} g_{w(\alpha)_v, w(\beta)_v} - \left(\sum_{i,j} g_{ij}\right)^m. \tag{8}$$

Indeed $\langle W_\alpha, W_\beta \rangle$ is the displayed product. Writing $s_i = \sum_j g_{ij}$, the multinomial theorem gives $\langle F_m, G_m \rangle = (\sum_i s_i)^m/(m!)^2 = \|G_m\|_{\mathrm{HS}}^2$, which proves (8).

Response bound. For $X_i = R_A(U_i)$ on the unit tuple ball, put

$$M = \sup_A \sum_i \|X_i\|_{\mathrm{op}}, \qquad K = \sup_A \sum_{i<j} \|[X_i, X_j]\|_{\mathrm{op}}.$$

If $M = 0$, all block responses vanish. For chains the positive certificate bound below proves separation directly; otherwise assume $M > 0$. The usual two-factor commutator representation [7] gives

$$\left\| e^{\lambda X} e^{\lambda Y} - e^{\lambda(X+Y)} \right\|_{\mathrm{op}} \leq \frac{\lambda^2}{2} \|[X, Y]\|_{\mathrm{op}}\, e^{\lambda(\|X\|_{\mathrm{op}} + \|Y\|_{\mathrm{op}})}.$$

For completeness, differentiate $e^{(\lambda-s)(X+Y)} e^{sX} e^{sY}$ and use

$$[e^{sX}, Y] = \int_0^s e^{(s-u)X} [X, Y] e^{uX}\, du.$$

Induction over factors, with $[\sum_{i<j} X_i, X_j] = \sum_{i<j} [X_i, X_j]$, yields

$$\Delta_n(\lambda) \leq \frac{\lambda^2 K}{2} e^{\lambda M}.$$

Comparing power coefficients with $\mathcal{B}(\lambda) = \sum_{m \geq 2} \lambda^m H_m/m!$, a sufficient condition is

$$K < \inf_{m \geq 2} \frac{2H_m}{m(m-1)M^{m-2}} \quad (M > 0).$$

This infimum is genuinely infinite.

Analytic verification for chains. For $U_i = [e_i, e_{i+1}]$, blocks are mutually orthogonal with norm $\sqrt{2}$. Let $a_i = \|A_i\|_F$ and $c_\ell = \cos(\pi/(\ell+2))$. The path-graph largest eigenvalue gives $\sum_i a_i a_{i+1} \le c_\ell \sum_i a_i^2 \le c_\ell$. Böttcher–Wenzel therefore gives

$$M \le \sqrt{2} c_\ell < \sqrt{2}, \qquad K \le 2\sqrt{2} \sum_{i<j} (a_i a_{i+1})(a_j a_{j+1}) \le \sqrt{2} c_\ell^2 < \sqrt{2}.$$

The K inequality is pointwise before taking its supremum. Retain only nonmonotone words over the first three orthogonal blocks. There are $3^m - \binom{m+2}{2}$ such words, each absent from F_m and having coefficient $1/m!$ in G_m. Hence

$$H_m^2 \ge 2^m (3^m - \binom{m+2}{2}) \ge 2^m 3^{m-1}, \qquad m \ge 2.$$

The counting inequality follows at $m = 2$ and by induction: $\binom{m+3}{2}/\binom{m+2}{2} = (m+3)/(m+1) < 3$. Consequently every coefficient threshold is at least

$$L_m = \frac{4\, 3^{(m-1)/2}}{m(m-1)}.$$

Its consecutive ratio is $\sqrt{3}(m-1)/(m+1)$: below one up to $m = 3$, and above one for $m \ge 4$. For $m = 4 + s$, the latter follows from $3(m-1)^2 - (m+1)^2 = 2s^2 + 8s + 2 > 0$. Thus $\inf_{m \ge 2} L_m = L_4 = \sqrt{3} > \sqrt{2} > K$. Summing $H_m \ge (\sqrt{6})^m/\sqrt{3}$ gives the last bound in (5); the same coefficient comparison gives its middle strict inequality. This proves Theorem 2 at every positive scale.

C Euler minimax proof

Put $x_i = |A|\, \|X_i\|_{1\text{-var}}$ and $R = \sum_i x_i$. Expanding all exact interval factors, the Euler product retains precisely multi-indices with every interval degree at most N. Factorial estimates give

$$\left\| \Phi^A(X) - E^A_{N,\pi}(X) \right\|_{\mathrm{op}} \le \sum_{\max j_i > N} \prod_i \frac{x_i^{j_i}}{j_i!} = e^R - \prod_i P_N(x_i).$$

The majorant increases in each x_i. For $N \ge 2$, direct coefficient comparison gives

$$P_{N-1}(x)^2 - P_N(x) P_{N-2}(x) = \frac{x^{N-1}}{N!} \sum_{j=0}^{N-1} (N-j) \frac{x^j}{j!} \ge 0.$$

For $N = 1$, $\log P_1 = \log(1+x)$ is concave as well. Thus $\log P_N$ is concave, and Jensen maximizes the product at equal $x_i = R/m$. An equal-variation partition is independent of A and gives the uniform bound $\mathcal{E}(N, m, r)$ on the normalized path class.

For $X_t = te_1$, take $A_1 = rQ$ with a rank-one orthogonal projection and other matrices zero. On its range the error is $e^r - \prod_i P_N(r\Delta t_i)$. Jensen shows every partition has error at least $\mathcal{E}(N, m, r)$; equal intervals attain it. This proves the quantifier order in (6).

D Storage limit arguments

Write $x = z/m$, $a = e^{-x} \sum_{j>q} x^j/j!$, and $B = e^z z^{q+1}/((q+1)! m^q)$. Poisson-tail integral bounds, or repeated integration of e^{-u}, give

$$e^{-x} \frac{x^{q+1}}{(q+1)!} \le a \le \frac{x^{q+1}}{(q+1)!}.$$

Since $\mathcal{E} = e^z[1-(1-a)^m]$, Bernoulli's inequality yields

$$\frac{e^{-x}}{1+ma} \le \frac{\mathcal{E}}{B} \le 1. \tag{9}$$

Feasibility $\mathcal{E} \le \varepsilon < e^z$ and $1-(1-a)^m \ge 1-e^{-ma}$ also imply

$$ma \le -\log(1-\varepsilon e^{-z}). \tag{10}$$

Optimizer regime. For fixed d, k, Witt's formula gives $\log \Lambda_d(kq) = \beta q - \log q + O(1)$. Take q nearest $\sqrt{\lambda/\beta}$ and

$$m = \left\lceil z \exp\left(\frac{\lambda - \log((q+1)!)}{q}\right)\right\rceil.$$

This is B-feasible. Stirling and negligible relative rounding give $\log(C_{d,k}/z) \le 2\sqrt{\beta\lambda} + O(\log\lambda)$. Because $m \ge 1$, every optimizer has $q = O(\sqrt{\lambda})$. In the large-z regime, $\log z = O(\log\lambda)$ in this inference.

At fixed z, bounded m and $q = O(\sqrt{\lambda})$ would, by the first omitted term and $1-(1-a)^m \ge a$, give $\log\mathcal{E} \ge -O(\sqrt{\lambda}\log\lambda)$, contradicting $\log\varepsilon = -\lambda + O(1)$. Thus $m \to \infty$ and $x \to 0$. At $z \to \infty$, $m = 1$ gives $\log P_q(z) = O(\sqrt{z}\log z) = o(z)$, so $\mathcal{E} \sim e^z$, infeasible. If $m \ge 2$ and $x \ge c > 0$, then

$$\mathcal{E} \ge e^{z-x}\frac{x^{q+1}}{(q+1)!} \quad\Longrightarrow\quad \log\mathcal{E} \ge z/2 - O(\sqrt{z}\log z),$$

also infeasible. Hence $x \to 0$, and $m \to \infty$, in both regimes.

At fixed z, $ma \le zx^q/(q+1)! \le zx/2 \to 0$ once $x < 1$. For large z, (10) gives $ma \to 0$. It follows from (9) that $\mathcal{E}/B \to 1$ at every optimizer, so $\log B \le \log\varepsilon + o(1)$. Consequently

$$\log(C_{d,k}/z) \ge \beta q + \lambda/q - O(\log\lambda).$$

The minimum of $\beta q + \lambda/q$ is $2\sqrt{\beta\lambda}$. Its gap from that minimum forces $q/\sqrt{\lambda/\beta} \to 1$. Subtracting $\log\Lambda_d(kq)$ gives the formula for $\log(m/z)$.

Finally, at an exact optimizer inflate m by $\max\{1, (B/\varepsilon)^{1/q}\}$, rounding upward. This is B-feasible and costs a relative $1+o(1)$ since $B/\varepsilon \le 1+o(1)$ and $m \to \infty$. The exact feasible region contains the B-feasible one, proving convergence of the cost ratio to one.

E Engel gauge and structured storage scope

Repeated Böttcher–Wenzel gives

$$\left\|\mathrm{ad}_A^{k-1}(B)\right\|_{\mathrm{op}} \le 2^{(k-1)/2}\,\|A\|_F^{k-1}\,\|B\|_F\,.$$

Under $\|A\|_F^2 + \|B\|_F^2 \le 1$, the maximum occurs at squared norms $(k-1)/k$ and $1/k$, giving g_k. A diagonal block $A = \sqrt{(k-1)/(2k)}\,\mathrm{diag}(1,-1)$ and $B = E_{12}/\sqrt{k}$ attain operator norm g_k. Binomial expansion of E_k gives HS norm h_k. Its ratio g_k/h_k decays exponentially in k; this is a sharp corollary of a classical commutator inequality.

For the pure weakly geometric path $\exp_\otimes(taE_k)$, use $p \ge k$. At Euler depth kq, coefficientwise nonnegative error terms give

$$\mathcal{E}(q,m,as_k) \le \delta_k(q,m,a) \le \mathcal{E}(q,m,ag_k) < \mathcal{E}(q,m,ah_k).$$

For the lower witness, set $\alpha = \sqrt{(k-1)/k}$, $\eta = 1/\sqrt{k}$:

$$A_1 = \frac{\alpha}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad A_2 = \frac{\eta}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ (-1)^{k-1} & 0 \end{pmatrix}.$$

Their unit tuple evaluates E_k to a symmetric matrix with off-diagonal entries s_k, hence positive eigenvalue s_k. Jensen gives the same witness lower bound for adaptive partitions. For the HS equality, each tensor-degree-kj error is $c_j a^j E_k^{\otimes j}$ with $c_j \geq 0$. Its norm is $c_j (a h_k)^j$, whose sum equals $\mathcal{E}(q, m, a h_k)$.

For $\varepsilon < e^{a s_k} - 1$, depths below k are infeasible; intermediate depths have the same polynomial as the preceding multiple of k and larger storage. This gives the stated storage sandwich. The amplitude corollary follows from (7). At fixed a, all bracket endpoints instead have common leading logarithmic rate $2\sqrt{k \log d \log(1/\varepsilon)}$.

F Rational certificates and reproducibility

For rational $x \geq 0$ and integer J with $x < J + 2$,

$$P_J(x) \leq e^x \leq P_J(x) + \frac{x^{J+1}}{(J+1)!} \frac{1}{1 - x/(J+2)}.$$

This follows by a geometric upper bound on all successive omitted-term ratios. All displayed example enclosures use $J = 24$. Integer squares verify

$$\sqrt{2} < 707107/500000, \quad \sqrt{3} > 34641/20000, \quad \sqrt{6} < 244949/100000.$$

For the corner, $\rho h = \sqrt{6}/27$. Positive coefficients permit replacing $\sqrt{2}$ parameters by their upper bounds in Q_R; apply the exponential enclosure, and use the lower $\sqrt{3}$ bound in ht^3. Exact fractions then give (4).

For the storage example, bound $e = e^1$ by the same rational enclosure. The first five cumulative dimensions are $2, 3, 5, 8, 14$. Enumerating every (N, m) of cost at most 8 for $\mathcal{E}$, and at most 14 for B, gives the two unique optima in the table. Higher depths already cost more with $m = 1$. Each feasibility inequality is strictly decided by the rational bounds.

Source and computed examples are shipped in the accompanying source archive. Versioned source comprises the response modules, tests, runners and notebooks. The PDF is self-contained. The grid statistics are regenerated from retained CSV and checked against current integer optimization by the build runner. Ordinary floating-point plots/searches have no interval-certification claim.

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